

README

HelixCode

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Containerized, network-exposed, browser-based **VS Code** (code-server) that mounts your host projects for immediate work — behind a **Caddy** TLS edge, running **rootless** (Podman), surviving reboot via a systemd service.

Release codeserver-1.0.0-dev-0.0.3 (2026-07-01). Authentication pivots to the **real-account, SSH-key challenge-response** model — a session runs host-native as the real host user (milosvasic) with that user's ~/.ssh, .bashrc, and host binaries, and login signs a fresh server-issued challenge against the account's authorized_keys (**no password, nothing stored**). The single CODE_SERVER_PASSWORD login is **retired**. New model + prerequisites + troubleshooting: [docs/guides/AUTH.md](#); full change list: [docs/changelogs/codeserver-1.0.0-dev-0.0.3.md](#). **In progress — not yet validated** (§11.4.6); see [docs/features/Status.md](#).

Quick start (reproducible, reboot-persistent)

```
scripts/install.sh  
# preflight → wizard (if needed) → start → boot service
```

Then open **https://<host-ip>:52443** and log in with your **host account** (milosvasic) SSH key — you sign a fresh challenge shown on the login page, no password, nothing stored (see [Authentication](#) below). [http://<host-ip>:52080](#) redirects to HTTPS.

Manual flow: `scripts/doctor.sh` → `scripts/setup.sh` → `scripts/start.sh` → `scripts/status.sh`. Full script reference: [docs/scripts/README.md](#).

What you get

- **Caddy edge** on 0.0.0.0:52443 (HTTPS) / :52080 (HTTP→HTTPS), reverse proxy to code-server. Ports derive from PORT_PREFIX in deploy/.env.
- **code-server** serving your host projects **read-write** (bind-mounted).
- **Real-account sessions** (*new, in progress*) — the editor runs host-native **as the real user**, so it natively has that user's SSH keys, .bashrc, and all host binaries; login is an **SSH-key challenge-response** verified against the account's `authorized_keys` — no password, nothing stored. See [Authentication](#).
- **Rootless** containers (Podman, §11.4.161) — no root, no privilege escalation (Caddy TLS edge; the editor + auth run host-native).
- **Boot-survival** rootless systemd user service (linger enabled).
- **File-watcher fix** — `files.watcherExclude` seeded into a persistent volume so large trees don't hit the inotify limit.

Authentication (real account + SSH-key)

New model, in progress — not yet validated. This replaces the old single `CODE_SERVER_PASSWORD` login (and supersedes the earlier PAM-login draft — non-root PAM verify is impossible on this ALT/tcb host, so auth pivoted to SSH keys). The components are being built and their tests/evidence are still being produced; treat the behavior below as *intended*, not yet proven. Full guide: [docs/guides/AUTH.md](#).

Each session is tied to a **real host user account** (default milosvasic). The editor runs **host-native as that user**, so it natively inherits the user's **SSH keys** (`~/.ssh` — git over SSH works from the integrated terminal), full **.bashrc/profile**, and **all installed host binaries**. The login screen shows a **fresh signed challenge**; you sign it locally with your SSH key —

```
printf %s '<challenge>' | ssh-keygen -Y sign -n helixcode-login  
-f ~/.ssh/id_ed25519
```

— and paste the signature back. A non-root helix-auth forward-auth gate behind Caddy **verifies it against the account's `authorized_keys`** (`ssh-keygen -Y verify`) and issues a signed session cookie. It **stores nothing** — no password anywhere (repo, config, env, logs, evidence); your private key never leaves your machine.

The editor's file **Explorer defaults to `$PROJECTS_ROOT`** for convenience — it is **NOT a security boundary**: the integrated terminal, **File > Open Folder**, and extensions retain **full real-**

user (milosvasic) access to the host filesystem BY DESIGN. Isolation, if ever required, would need a container / VM / chroot (out of scope per operator decision). Defense-in-depth (deploy stream, in progress): code-server runs with `--disable-workspace-trust` and the `/proxy/` path is blocked at the Caddy edge.

New parameters (replacing `CODE_SERVER_PASSWORD`, which is retired):

```
HELIX_AUTH_MODE=sshkey           # only mode this
                                  release
HELIX_AUTH_ACCOUNT=milosvasic    # the host account
                                  sessions tie to
HELIX_AUTH_AUTHORIZED_KEYS=~/.ssh/authorized_keys # verifier's
                                  trust source
HELIX_AUTH_PRINCIPAL=milosvasic  # expected signer
                                  principal
PROJECTS_ROOT=                   # workspace +
                                  default Explorer folder (not a jail)
# No password parameter – login signs a server-issued challenge,
  verified against authorized_keys.
```

Security posture: auth **fails closed** (if the helix-auth gate is down, access is denied — never bypassed), the challenge is **server-issued + short-lived**, the session cookie is **HMAC-signed + HttpOnly + Secure + SameSite**, and login is **rate-limited**. Details, prerequisites, and troubleshooting: <docs/guides/AUTH.md>.

HTTPS / TLS

Default is a per-boot **self-signed** cert (LAN-friendly). For real **Let's Encrypt** with automatic renewal + rotation, set `TLS_MODE=letsencrypt` (+ `CS_DOMAIN`, `ACME_EMAIL`) in `deploy/.env` — you need a public domain resolving to the host with reachable `:80/443`, **or** a DNS-01 token. Modes, prerequisites, renewal/rotation behavior, and the local-CA proof: <docs/guides/TLS.md>.

Testing (anti-bluff, §11.4.169)

```
bash tests/run_all_types.sh      # full test-type matrix
                                  (needs the stack)
bash tests/run_all_types.sh --list # suites in risk-descending
                                  order
```

14 suites under `tests/types/` — unit, integration, e2e, full-automation, security, load (DDoS), stress+chaos, concurrency, race, memory, benchmark, Let's-Encrypt (local ACME CA), Challenges, HelixQA. Every PASS cites a captured evidence file (`qa-results/`). Feature ledger: <docs/features/Status.md>.

Governance

Governed by the **Helix Constitution** (constitution/) — anti-bluff covenant, rootless containers, no force-push, project-prefixed release tags. See [CLAUDE.md](#).